LECTURE PLAN

Institute/Department	UNIVERSITY INSTITUTE OF COMPUTING (UIC)	**Program**	**Master of Computer Applications - Artificial Intelligence & Machine Learning(MC306)**
Master Subject Coordinator Name:	Sameera Dhuria	**Master Subject Coordinator E-Code**:	**E17674**
Course Name	Design and Analysis of Algorithms	**Course Code**	**25CAT-611**

Lecture	Tutorial	Practical	Self Study	Credit	Subject Type
3	0	0	0	3.00	T

Course Type	Course Category	Mode of Assessment	Mode of Delivery
Major Core	Graded (GR)	Theory Examination (ET)	Theory (TH)

Mission of the Department	M1 To provide innovative learning centric facilities and quality-oriented teaching learning process for solving computational problems. M2 To provide a frame work through Project Based Learning to support society and industry in promoting a multidisciplinary activity. M3To develop crystal clear evaluation system and experiential learning mechanism aligned with futuristic technologies and industry. M4 To provide doorway for promoting research, innovation and entrepreneurship skills in collaboration with industry and academia. M5 To undertake societal activities for upliftment of rural/deprived sections of society
Vision of the Department	To be a Centre of Excellence for nurturing computer professionals with strong application expertise through experiential learning and research for matching the requirements of industry and society instilling in them the spirit of innovation and entrepreneurship.

Program Educational Objectives(PEOs)	
PEO1	Demonstrate analytical and design skills including the ability to generate creative solutions and foster team-oriented professionalism through effective communication in their careers.
PEO2	Expertise in successful careers based on their understanding of formal and practical methods of application development using the concept of computer programming languages and design principles in accordance to industry 4.0
PEO3	Exhibit the growth of the nation and society by implementing and acquiring knowledge of upliftment of health, safety and other societal issues.
PEO4	Implement their exhibiting critical thinking and problem- solving skills in professional practices or tackle social, technical and business challenges.

Program Specific OutComes(PSOs)	
PSO1	Analyze their abilities in systematic planning, developing, testing and executing complex computing applications in field of Social Media and Analytics, Web Application Development and Data Interpretations.
PSO2	Apprise in-depth expertise and sustainable learning that contributes to multi-disciplinary creativity, permutation, modernization and study to address global interest.

Program OutComes(POs)	
PO1	Apply mathematics and computing fundamental and domain concepts to find out the solution of defined problems and requirements.
PO2	Use fundamental principle of Mathematics and Computing to identify, formulate research literature for solving complex problems, reaching appropriate solutions.
PO3	Understand to design, analyse and develop solutions and evaluate system components or processes to meet specific need for local, regional and global public health, societal, cultural, and environmental systems.
PO4	Use expertise research-based knowledge and methods including skills for analysis and development of information to reach valid conclusions.
PO5	Adapt, apply appropriate modern computing tools and techniques to solve computing activities keeping in view the limitations
PO6	Exhibiting ethics for regulations, responsibilities and norms in professional computing practices.

PO7	Enlighten knowledge to enhance understanding and building research, strategies in independent learning for continual development as computer applications professional.
PO8	Establishing strategies in developing and implementing ideas in multi- disciplinary environments using computing and management skills as a member or leader in a team.
PO9	Contribute to progressive community and society in comprehending computing activities by writing effective reports, designing documentation, making effective presentation, and understand instructions
PO10	Apply mathematics and computing knowledge to access and solve issues relating to health, safety, societal, environmental, legal, and cultural issues within local, regional and global context.
PO11	Gain confidence for self and continuous learning to improve knowledge and competence as a member or leader of a team.
PO12	Learn to innovate, design and develop solutions for solving real life business problems and addressing business development issues with a passion for quality competency and holistic approach.

Text Books

Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Introduction to the Design and Analysis of Algorithms	Anany Levitin	2nd Edition	Pearson	2009
2	Computer Algorithms	Sartaj Sahni and Rajasekaran	Volume 1	Computer Science Press	1997

Reference Books

Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronald L.	3rd Edition	PHI (Prentice Hall of India)	2009
2	Design and Analysis of Algorithms	S. Sridhar	Edition 3	Oxford University Press (Higher Education division	2007

Course OutCome

SrNo	OutCome
CO1	Understand the basics of different data structures to manage the data
CO2	Analyze the asymptotic performance of algorithms through algorithmic complexity of simple, non-recursive programs.
CO3	Understand the fundamentals of data structures
CO4	Apply and analyze important algorithmic design paradigms and their applications.
CO5	Implement the major graph algorithms to model engineering problems.

Lecture Plan Preview-Theory

Unit No	LectureNo	ChapterName	Topic	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Numer (s)
1	1	Introduction	Introduction to Algorithms and Their Characteristics	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO1
1	2	Introduction	Algorithm Specification and Problem Solving Approach	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO1
1	3	Introduction	Algorithm Analysis Framework and Performance Metrics	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO2

1	4	Introduction	Space and Time Complexity Concepts	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO2
1	5	Introduction	Asymptotic Notations: Big-O, Omega, Theta, and Little-oh –	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO2
1	6	Introduction	Mathematical Analysis of Non-Recursive Algorithms with Examples	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO2
1	7	Introduction	Mathematical Analysis of Recursive Algorithms with Examples	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO2
1	8	Introduction	Linear and Binary Search Algorithms	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO1
1	9	Introduction	Sorting Algorithms: Bubble Sort and Quick Sort	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO1
1	10	Introduction	Merge Sort and Complexity Comparison – CO2	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO2
1	11	Introduction	Linked Lists – Single, Double, and Circular	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO1
1	12	Introduction	Operations on Linked Lists: Traversing, Insertion, and Deletion	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO1
1	13	Introduction	Stacks and Queues: Concepts and Applications	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO1
1	14	Introduction	Graphs: Representation and Basic Operations	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO1
1	15	Introduction	Trees and Binary Search Trees	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO1
2	16	Algorithm Design Paradigm	General Method of Divide and Conquer	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
2	17	Algorithm Design Paradigm	Advantages and Disadvantages of Divide and Conquer	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
2	18	Algorithm Design Paradigm	Introduction to Decrease and Conquer Approach	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
2	19	Algorithm Design Paradigm	Topological Sort using Decrease and Conquer –	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4

2	20	Algorithm Design Paradigm	Divide and Conquer vs Decrease and Conquer	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
2	21	Algorithm Design Paradigm	Fractional Knapsack Problem	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
2	22	Algorithm Design Paradigm	Introduction to Graph Algorithms	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO3
2	23	Algorithm Design Paradigm	Prim’s Algorithm for Minimum Spanning Tree	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
2	24	Algorithm Design Paradigm	Kruskal’s Algorithm for Minimum Spanning Tree	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
2	25	Algorithm Design Paradigm	Comparison of Prim’s and Kruskal’s Algorithms	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
2	26	Algorithm Design Paradigm	Single Source Shortest Path Problem	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
2	27	Algorithm Design Paradigm	Dijkstra’s Algorithm	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
2	28	Algorithm Design Paradigm	Introduction to Huffman Trees and Codes	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
2	29	Algorithm Design Paradigm	Heap Data Structure	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO3
2	30	Algorithm Design Paradigm	Heap Sort Algorithm	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
3	31	Dynamic Programming	General Method with Examples	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
3	32	Dynamic Programming	Multistage Graphs	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO5
3	33	Dynamic Programming	Binomial Coefficient	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
3	34	Dynamic Programming	Warshall’s Algorithm	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO5
3	35	Dynamic Programming	All Pairs Shortest Path: Floyd’s Algorithm	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO5

3	36	Dynamic Programming	Optimal Binary Search Trees	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
3	37	Dynamic Programming	0/1 Knapsack Problem	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
3	38	Dynamic Programming	Bellman-Ford Algorithm	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO5
3	39	Dynamic Programming	Travelling Salesperson Problem (TSP)	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
3	40	Dynamic Programming	N-Queens Problem	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
3	41	Dynamic Programming	Sum of Subsets Problem	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4
3	42	Dynamic Programming	Graph Coloring	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO5
3	43	Dynamic Programming	Hamiltonian Cycles	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO5
3	44	Dynamic Programming	Assignment Problem	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO5
3	45	Dynamic Programming	P, NP, NP-Hard, NP-Complete, Co-NP, LC Branch and Bound	,T-Computer Algorithms,T-Introduction to the Design and,R-Design and Analysis of Algorit,R-Introduction to Algorithms	PPT	CO4

Assessment Model

Sr No	Assessment Name	Exam Name	Max Marks
1	20EP02	External Theory	60
2	20EP02	Attendance Marks	2
3	20EP02	Mid-Semester Test-1	20
4	20EP02	Short Term Paper / Research Paper	12
5	20EP02	Mid-Semester Test-2	20
6	20EP02	Quiz	6

CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	NA	NA	1	1	NA	NA	NA	NA	NA	NA	NA	NA	1
CO2	3	NA	3	2	2	NA	NA	NA	NA	NA	NA	NA	1	1
CO3	3	NA	3	2	3	NA	NA	NA	NA	NA	NA	NA	1	1
CO4	2	3	NA	1	2	NA	NA	NA	NA	NA	NA	NA	1	3
CO5	2	NA	2	2	3	NA	NA	NA	NA	NA	NA	NA	3	2
Target	2.6	3	2.67	1.6	2.2	NA	NA	NA	NA	NA	NA	NA	1.5	1.6